

DATA ANALYSIS PROJECT REPORT

Project Title	Highway Traffic Data Analysis
Student Name	MD Mazharul Islam Nabil
Student ID	23-50025-1
Course / Section	Programming in Python, section D
Instructor	MD. TANZEEM RAHAT
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1. Executive Summary

This project analyzes the "Metro_Interstate_Traffic_Volume.csv" dataset to determine how the time of day, adverse weather conditions, and the day of the week impact the number of vehicles on the I-94 highway. The primary goal was to uncover predictable traffic patterns and identify anomalies using a structured data analysis pipeline. Using Python, the dataset was cleaned to remove invalid entries and duplicates, while feature engineering was applied to extract time-based and weather-based indicators. Methodologies involved using Pandas for grouping and aggregation, NumPy for variance and Z-score anomaly detection, and Matplotlib for visualizing trends. The strongest findings reveal a clear bimodal "rush hour" pattern peaking at 7 AM and 4 PM, a significantly higher traffic volume on weekdays (averaging 3,408 vehicles) compared to weekends (2,371 vehicles), and the presence of 17 extreme midnight traffic anomalies that defy standard daily trends.

2. Introduction and Project Objectives

Understanding highway traffic patterns is critical because traffic jams cost money, pollute the air, and damage roads. By analyzing the volume of cars on the I-94 highway, we can better understand how daily commutes and environmental factors influence congestion.

Item	Your Content
Dataset name	Metro Interstate Traffic Volume
Dataset source	https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume
Main project goal	To evaluate how time of day, bad weather, weekdays, and weekends change the number of traffic on the I-94 highway.
Research Question 1	Exactly when is "Rush Hour," and how bad does it get?
Research Question 2	Does extreme bad weather (like heavy snow or freezing cold) keep cars off the road?

Research Question 3	How does traffic change between weekdays and weekends?
Research Question 4	Can we use math to find weird traffic jams, like a jam at midnight?

3. Dataset Description

The dataset contains hourly recordings of traffic volume alongside weather and time-based metrics. Each row represents a specific hour's traffic flow and environmental conditions.

3.1 Basic Dataset Information

Property	Value
Number of rows	48,204 initially (48,177 after cleaning)
Number of columns	9 initially (15 after feature engineering)
Data types present	numerical (float64, int64) and categorical/date-time (object)
Target or key variable(s)	traffic_volume
Potential issues observed at first glance	Missing values in the 'holiday' column (48,143 missing cells), duplicate rows, and impossible temperature readings (0 Kelvin).

3.2 Initial Observations

Upon initial inspection, the 'holiday' column was almost completely empty, containing 48,143 missing values. The 'date_time' column required conversion from a string object to a datetime format for time-series analysis. Additionally, summary statistics revealed that the minimum temperature was 0.0 Kelvin, representing a physical impossibility for the highway's location.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Inconsistent column naming formats	Converted all column names to lowercase and stripped whitespace	Standardizes the data structure for easier and error-free indexing.	<code>df.columns = df.columns.str.strip().str.lower()</code>
Duplicate rows	Removed 17 duplicate rows	Prevents double-counting of identical hourly recordings,	<code>df = df.drop_duplicates()</code>

		which would skew volume averages.	
Impossible temperature readings	Removed 10 rows where temperature was ≤ 0 Kelvin	0 Kelvin is absolute zero, representing sensor errors rather than actual weather.	<code>df = df[~invalid_mask].copy()</code>
Missing values in the holiday column	Dropped the 'holiday' column entirely	The column contained almost exclusively missing values, making it analytically useless.	<code>df = df.drop(columns=['holiday'])</code>

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
temp_celsius	Subtracted 273.15 from the 'temp' column.	Converting temperature from Kelvin to Celsius makes the data easier to read and helps interpret how temperature affects traffic.
hour, month, day_name	Extracted directly from the 'date_time' datetime object.	These help identify traffic patterns at different times of the day, detect peak traffic hours, and compare traffic across different days of the week.
is_weekend	Boolean flag checking if <code>date_time.dt.dayofweek >= 6</code> .	Makes it easier to separate and compare traffic patterns between working days and holidays.
is_rush_hour	Flagged hours 7-10 and 15-20, strictly filtering out weekends.	Helps identify whether a specific time belongs to peak traffic hours and makes traffic analysis more meaningful for congestion detection.

6. Analysis Methodology

- Pandas was used for loading the CSV, dropping missing or erroneous data, converting datetime strings, and aggregating data using `.groupby()` to calculate mean traffic volumes by weather type, day of the week, and rush hour status.

- NumPy was used for advanced mathematical anomaly detection, specifically generating a Z-score for traffic volume to isolate extreme high-traffic events, and calculating the variance (np.var) to contrast traffic stability between weekdays and weekends.
- Matplotlib was used to render correlation heatmaps, hourly trend line charts, scatter plots for temperature interactions, and bar charts for daily and weather-based traffic volume. [Replace the example bullet points above with your own brief methodology explanation if needed.]

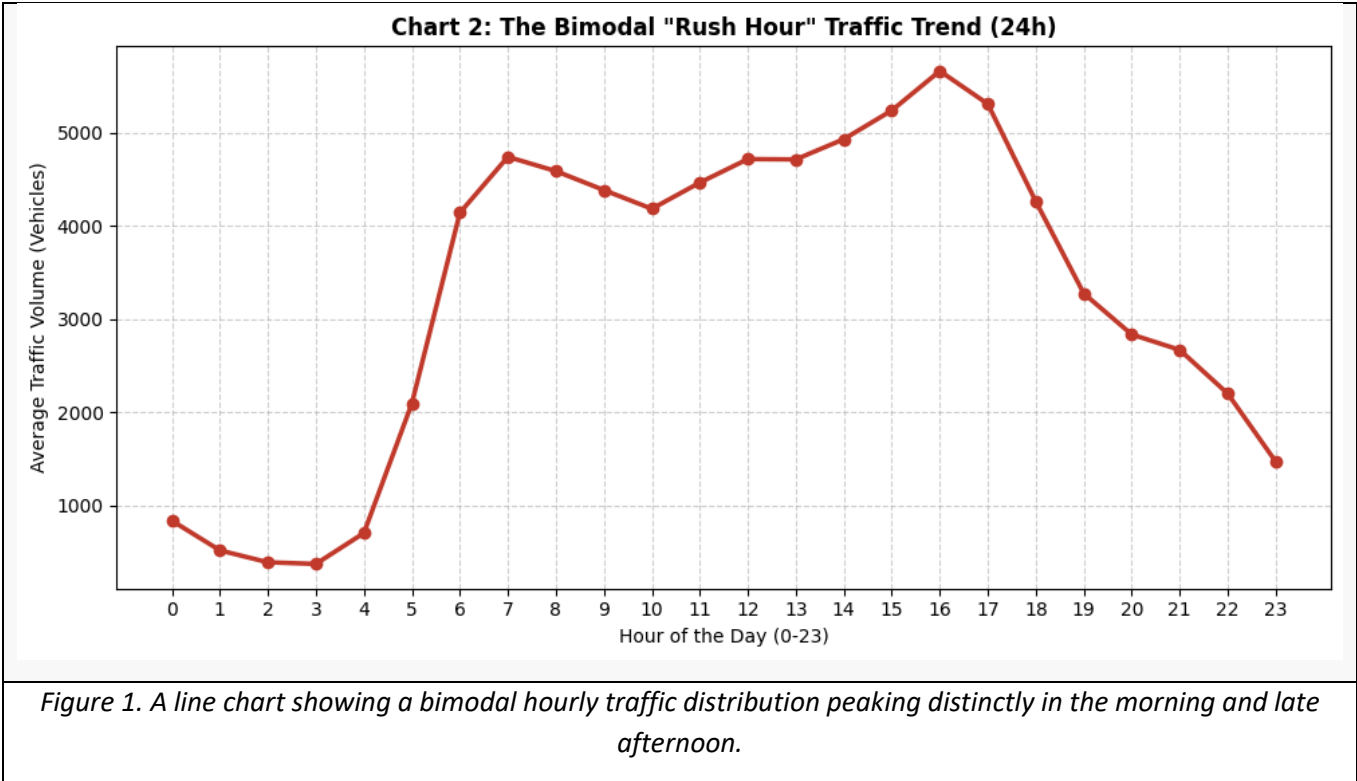
7. Analysis and Visualizations

[This is the main body of the report. Organize it by research question, not by code order. For each subsection, include the analysis, the relevant plot or table, and a short interpretation.]

7.1 Research Question 1

Exactly when is "Rush Hour," and how bad does it get?

Traffic volumes were grouped by the extracted 'hour' feature and averaged across the dataset. Furthermore, a dedicated is_rush_hour feature was calculated to compare peak periods against normal hours.

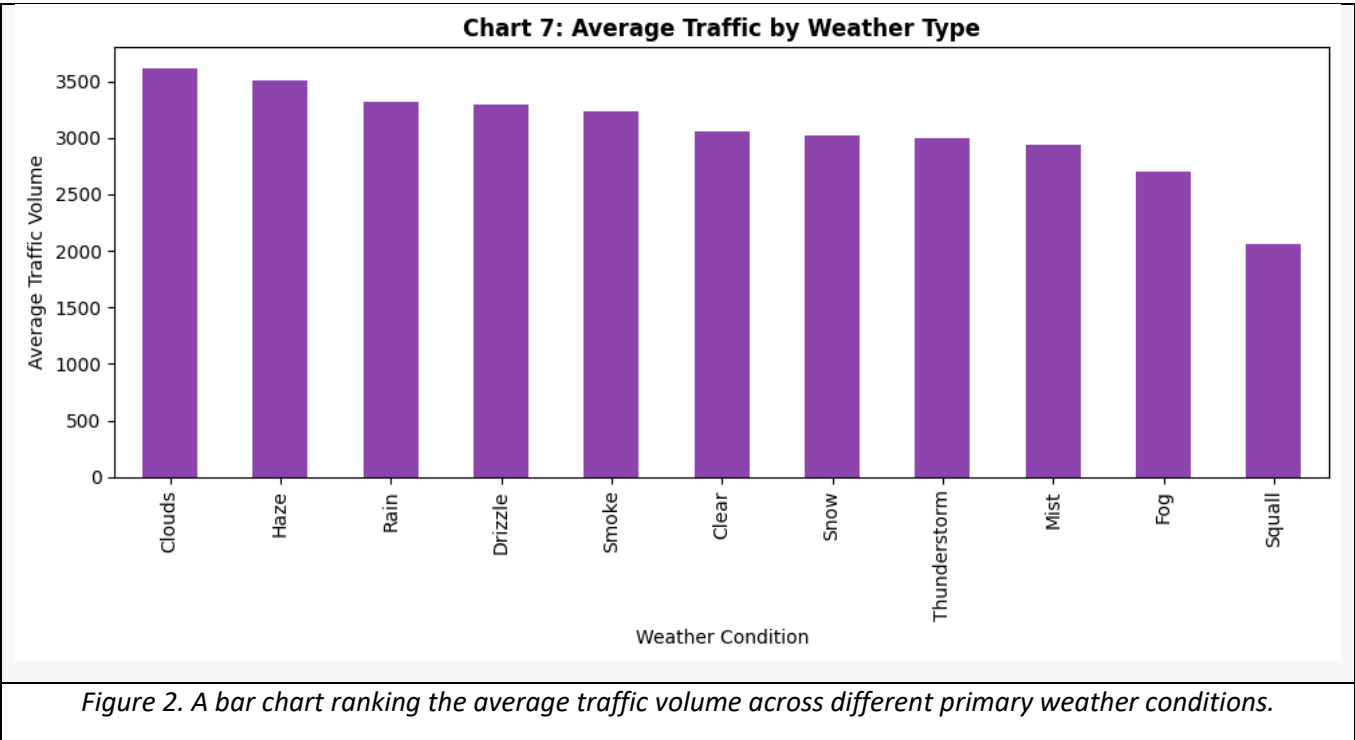


The analysis confirms a sharp bimodal traffic pattern, with clear peak hours occurring during morning commutes (7 AM) and evening commutes (4 PM / 16:00). During these flagged rush hours, the average volume surges to 4,673.57 vehicles, nearly double the non-rush hour average of 2,477.20 vehicles.

7.2 Research Question 2

Does extreme bad weather keep cars off the road?

Traffic was aggregated by the 'weather_main' category, and a Pearson correlation matrix was generated to measure mathematical relationships between numerical weather features (temp, rain, snow, clouds) and traffic volume.

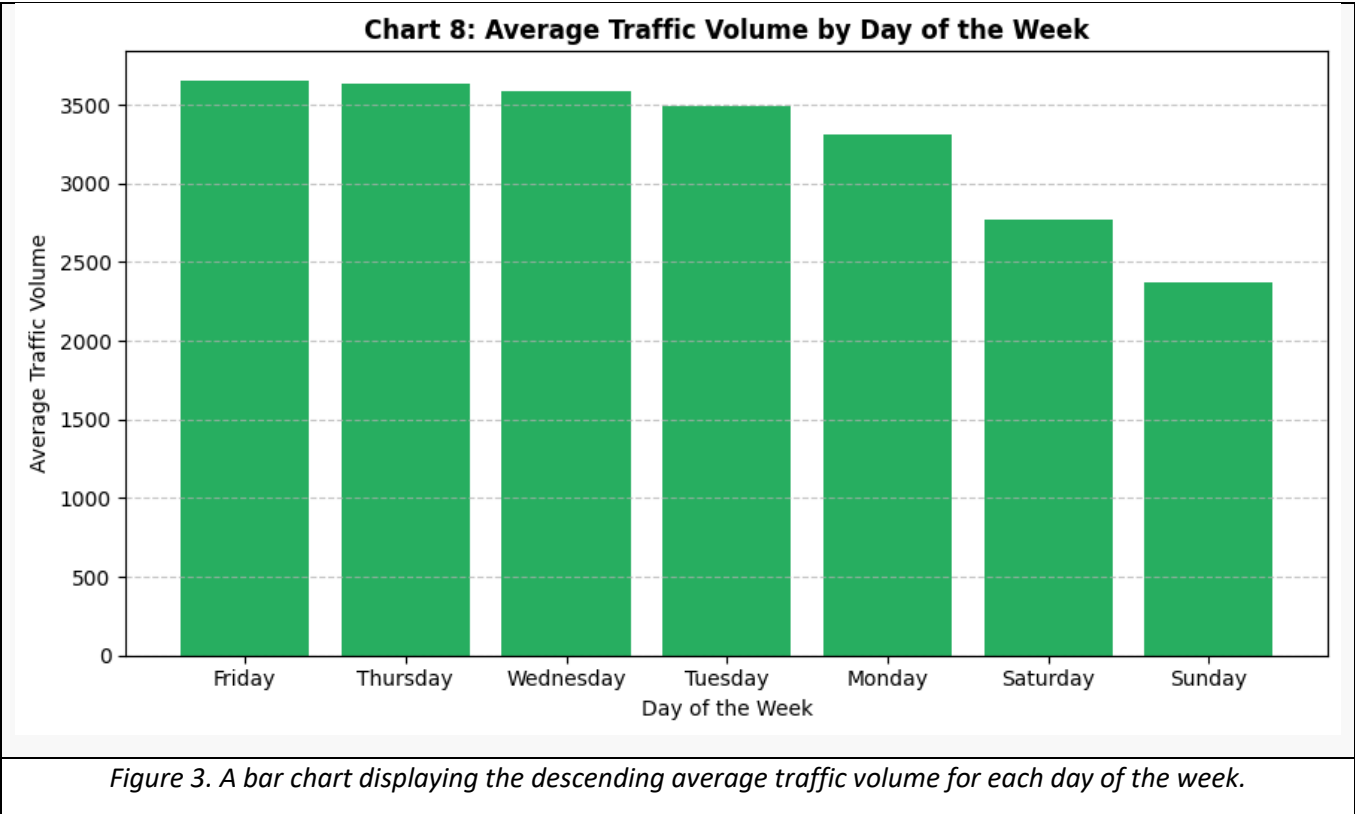


Weather conditions strongly affect traffic movement. Contrary to typical assumptions, normal weather like Clouds (3,617.98 avg volume) and Haze (3,502.10 avg volume) see the highest traffic, whereas extreme events like Squalls and Thunderstorms see reduced numbers. Furthermore, the correlation heatmap (Chart 1) shows a weak positive correlation (0.132) between Celsius temperature and traffic volume, indicating slightly higher traffic on warmer days.

7.3 Research Question 3

How does traffic change between weekdays and weekends?

Traffic volume was compared using the is_weekend flag and broken down granularly using the engineered day_name feature to calculate means. Variance was also calculated to see which period was more volatile.



Traffic behavior changes dramatically between working days and weekends. Weekdays maintain a high average volume of 3,407.81 vehicles, while weekends drop significantly to 2,370.53 vehicles. Friday represents the busiest day of the week (3,656.18 avg volume), while Sunday is the quietest. Additionally, weekday traffic is much more unstable, showing a variance of 4,085,134.48 compared to the weekend variance of 2,197,423.29.

8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, np.where-based classification, binning, or a custom score.]

Component	Description
Computation used	Traffic Volume Z-Score and Anomaly Detection.
Purpose	Z-score analysis helps identify extremely high or low traffic events compared to normal traffic conditions, which is useful for detecting unusual patterns.
Formula or logic	Calculated the mean (np.mean) and sample standard deviation (np.std(ddof=1)) of the traffic volume. The Z-score was generated using (df['traffic_volume'] - traffic_mean) / traffic_std.
Result / takeaway	By filtering for a Z-score > 2.0, the analysis successfully detected exactly 4 total extreme high-traffic events that deviate massively from the

	standard flow.
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9. Key Findings

- Finding 1: Traffic follows a strict bimodal trend dictated by working hours. Volume peaks specifically at 7 AM and 4 PM, nearly doubling the baseline hourly average.
- Finding 2: Weekdays carry significantly more traffic than weekends. Friday acts as the absolute peak for weekly volume, while Sunday is the lowest.
- Finding 3: Weekday traffic is highly variable. The variance for weekday traffic flow is roughly double the variance for weekend traffic, indicating more severe swings between empty roads and total gridlock.
- Finding 4: Extreme weather moderately suppresses traffic. Clearer conditions (Clouds/Haze) correlate with peak traffic, while adverse conditions (Squalls/Thunderstorms) align with lower traffic volumes.
- Finding 5: Midnight traffic anomalies occur. Despite standard volume dropping to near zero overnight, the data contains 17 distinct anomalies where midnight traffic spiked over 2,000 vehicles, likely indicating roadwork or late-night events.

10. Limitations

The primary limitation of this analysis is the removal of the holiday column. Because the feature contained 48,143 missing values, it had to be dropped, meaning we cannot definitively measure how national or regional holidays impact highway flow. Additionally, while we identified 17 midnight traffic anomalies, the dataset lacks contextual event data (e.g., concert schedules, accident reports, or construction logs) to explain *why* these specific jams occurred. Therefore, while we can prove mathematically that anomalies happen, we cannot determine causation.

11. Conclusion

The objective of this project was to understand the environmental and temporal factors influencing traffic congestion on the I-94 highway. The analysis successfully highlights that traffic is heavily dependent on the human work schedule, creating aggressive bimodal rush hours and drastically busier weekdays. Weather plays a secondary but measurable role, with extreme precipitation driving volumes down. Future analysis would benefit from incorporating localized event data or highway maintenance logs, which would allow us to decode the exact causes of the mathematical anomalies and midnight gridlocks discovered in this report.

12. References / Dataset Source

<https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume>

Final Submission Checklist

- I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- I attached the project report as a separate file.
- My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- My code uses Pandas, NumPy, and Matplotlib meaningfully.
- All bracketed placeholder text and helper notes were replaced or deleted before submission.